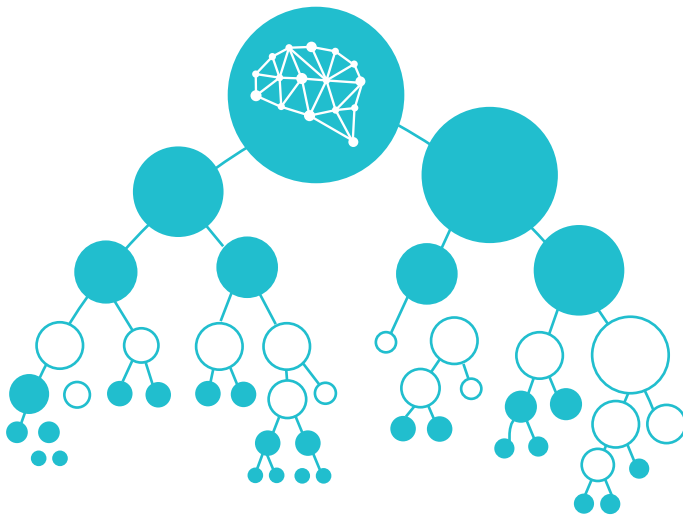


The Need for Ensemble Methods in Machine Learning

Ensemble methods train multiple models (base models) to solve the same problem and combine them for better performance. This offsets the individual variances and biases of the models. They can be used collectively for improved decision making. Let's find out how.



74 per cent accuracy. You may end up choosing a decision tree for this kind of classification, if you are only considering accuracy as one of the parameters. However, you may miss some of the important information that can be conveyed by the other models. Ensemble methods solve this problem; they combine different techniques together to give better accuracy and performance.

When ensemble methods are used, multiple models (base models) are trained to solve the same problem and combined together to get better performance. This offsets the individual variances and biases of the models. In simple terms, variances can be explained as how much the decisions made by individual models are different from each other. Bias is the difference between the predicted value of the model and the actual value.

A model is said to be performing well if it has a correct balance of low variance and low bias. If the bias is high, the model is underperforming and under-fitting, and if the variance is high it means the model is over-fitting. Over-fitting means the model tries to cover all or more than the required data points, and hence will not perform accurately in case of new unseen data, which will reduce its efficiency. Under-fitting means the model is not able to capture the exact trend of the data.

Ensemble methods, being a combination of techniques, tend to reduce variances by using something called bagging. Bagging reduces the bias by using 'boosting' and improves

It's a well-known fact that machine learning techniques can be broadly categorised into supervised, unsupervised and reinforcement learning. If data sets are labelled, supervised learning techniques can be used. If they are not labelled and we need to find some pattern in the data by analysing its characteristics, unsupervised learning techniques are used. Reinforcement learning works on the concept of reward and punishment to the model based on its interaction with the environment. This article reviews the ensemble methods in machine learning, which focus on supervised learning techniques.

Supervised learning can be broadly classified into classification and regression techniques. Classification is the supervised learning approach that categorises a given data into a set of classes. The popular classification techniques are – decision tree, logistic

regression, Naïve Bayes, support vector machine (SVM) and KNN. Regression is a supervised learning method that can predict a continuous-valued output for the new data given to the algorithm. Popular regression techniques include linear regression, polynomial regression, lasso regression, etc. The ensemble methods can be used for both regression and classification, without any alterations in the theoretical basis of the underlying algorithms. This helps them to address a broad range of machine learning problems.

Ensemble methods are an umbrella of machine learning algorithms that can be used collectively for improved decision making. Let's say you have a classification problem and you feed the same data to three different classifiers. Assume that the decision tree is giving 80 per cent accuracy, logistic regression is giving 75 per cent, and K nearest neighbor (KNN) is giving